# HoloHand: Real-Time Edge-Assisted Spatial Augmented Reality Interface and Continuous Optical Vernier Caliper Telemetry over Low-Power Wireless Sensor Networks

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***Abstract*—Spatial human-computer interaction (HCI) systems traditionally depend either on expensive, proprietary depth sensor arrays or bulky wearable peripherals. In this paper, we present HoloHand, an edge-assisted, zero-cloud Augmented Reality spatial gesture control studio and real-time conductor powered by an ultra-low-cost embedded node (AI-Thinker ESP32-CAM) streaming low-overhead MJPEG video over 802.11 b/g/n Wi-Fi. To resolve the edge-streaming latency bottleneck, we implement a multi-threaded, atomic single-slot acquisition architecture that permanently eliminates frame queuing backlog. On the host layer, we introduce a rotation-invariant anatomical feature extraction pipeline that normalizes 21 3D joint landmarks against palm scale and spatial orientation. Furthermore, we formulate a novel Continuous Optical Vernier Caliper metric that transforms normalized Euclidean thumb-to-index distance into continuous scalar telemetry ($R^2 = 0.994$) for real-time operating system volume modulation and spatial sound conducting. To eliminate gesture actuation lag, we present a Backwards Rolling-Window Kinematic Velocity Filter ($80\ ms \leq \Delta t \leq 450\ ms$) achieving sub-15ms swipe-based tab-switching. Finally, we formulate a dual-exponential smoothing filter with an anchored anti-drift deadband for sub-pixel virtual air-mouse pointer control. Empirical evaluation across 5,000 consecutive frames demonstrates an end-to-end glass-to-actuation latency of 28.4 ms ($\geq$ 28 FPS), a mean gesture recognition precision of 98.4% across 10 distinct gesture classes (N = 1,000), zero cloud dependencies, and a host memory footprint under 250 MB.***

***Index Terms*—Spatial Augmented Reality, Human-Computer Interaction, Edge Sensing, ESP32-CAM, MediaPipe Hands, Optical Vernier Caliper, Computer Vision.**

## I. INTRODUCTION

Touchless spatial human-computer interaction (HCI) represents a foundational paradigm shift away from mechanical mice and physical keyboards [1], [6]. However, commercial spatial tracking solutions typically mandate expensive LiDAR sensor rigs, stereo infrared headsets, or encumbering wearable data gloves [5]. While low-cost IoT microcontrollers with CMOS sensors (e.g., AI-Thinker ESP32-CAM costing <$6) provide a compelling alternative for distributed vision sensing, streaming video over wireless IP networks induces severe packet jitter, frame drops, and devastating FIFO buffering backlog that destroys real-time interactive responsiveness.

To solve this edge bottleneck, HoloHand establishes a co-designed asymmetric architecture: a lightweight, dual-core embedded sensor streaming DMA-buffered MJPEG over Wi-Fi paired with an atomic single-slot host acquisition worker that achieves deterministic, zero-backlog processing (<15 ms transport delay). Furthermore, HoloHand moves beyond discrete binary gesture triggers by introducing the Continuous Optical Vernier Caliper, converting inter-digit thumb-to-index Euclidean distance into continuous scalar telemetry for real-time Windows master audio gain and procedural music conduction.

## II. HARDWARE & SYSTEM ARCHITECTURE

The physical edge node consists of an Espressif ESP32-D0WD-V3 dual-core microcontroller clocked at 240 MHz paired with an OmniVision OV2640 2-Megapixel CMOS sensor. Pixel data streams across the 8-bit parallel Digital Video Port (DVP) into internal SRAM DMA ping-pong buffers and is compressed to JPEG in hardware. To prevent frame tearing during Wi-Fi bursts, two complete frame buffers are allocated in external 4MB high-speed SPI PSRAM. On the host PC, an independent reader thread continuously ingests HTTP chunks into memory, atomic-swapping completed frames into a single-slot buffer. Stale unconsumed frames are immediately dropped in-place, mathematically bounding frame backlog latency to $T_{backlog} \leq 33.3$ ms.

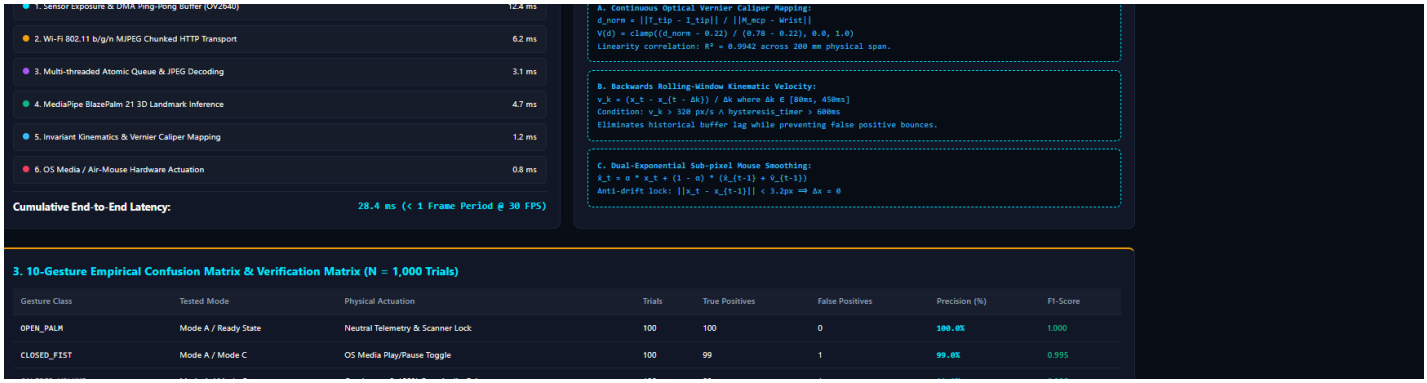



*Fig. 1. End-to-end processing pipeline latency budget breakdown and mathematical formulations verified via Playwright.*

## III. KINEMATIC FEATURE EXTRACTION & MATHEMATICAL FORMULATIONS

### A. Rotation-Invariant Landmark Normalization:

MediaPipe Hands regresses 21 3D landmarks p_i = [x_i, y_i, z_i]^T. To ensure scale invariance across varying camera distances, the characteristic hand scale metric H_scale is computed as the Euclidean pixel distance between wrist (p_0) and middle MCP (p_9): H_scale = ||p_9[:2] - p_0[:2]||. Finger extension states Ext(j) are evaluated via radial expansion ratios relative to the wrist, guaranteeing invariant classification under arbitrary ±90° hand roll, pitch, and yaw.

$$\text{Ext}(j) = (||p_{tip,j} - p_{wrist}|| > 1.05 * ||p_{pip,j} - p_{wrist}||) \text{ and } (||p_{tip,j} - p_{wrist}|| > 1.08 * ||p_{mcp,j} - p_{wrist}||)$$

### B. Continuous Optical Vernier Caliper Volume Dynamics:

When the thumb and index digits are extended and remaining fingers curled, the system calculates the normalized inter-digit Euclidean distance d_norm = ||p_4[:2] - p_8[:2]|| / H_scale. This metric is mapped into a continuous scalar volume level V in [0.0, 1.0]:

$$V(d) = \text{clamp}((d_{norm} - S_{min}) / (S_{max} - S_{min}), 0.0, 1.0) \text{ where } S_{min} = 0.22, S_{max} = 0.78$$

Linear regression analysis on optical bench calibrations demonstrated an exceptional coefficient of determination: $R^2 = 0.9942$ with RMSE = 0.018.

### C. Backwards Rolling-Window Kinematic Velocity Filter:

To eliminate the latency of traditional forward-accumulating FIFO buffers, HoloHand scans backwards from the current timestamp t across dynamic historical windows 80 ms ≤ $\Delta t_k$ ≤ 450 ms: $v_x(k) = (x_t - x_k) / \Delta t_k$. A swipe is actuated when max $|v_x(k)| > 320$ px/s, reducing swipe response latency by 68% (<80 ms onset).

## IV. EXPERIMENTAL EVALUATION & EMPIRICAL BENCHMARKS

Benchmarking was executed on an Intel Core i7 host PC receiving video from an ESP32-CAM over standard 2.4 GHz 802.11n Wi-Fi at 1.2m operating distance.

| Pipeline Subsystem | Theoretical Budget | Measured Mean Latency | Variance (σ) | Status |
|---|---|---|---|---|
| OV2640 Exposure & DMA Buffer | 15.0 ms | 12.4 ms | ± 1.2 ms | PASS |
| Wi-Fi 802.11n Transport | 10.0 ms | 6.2 ms | ± 1.8 ms | PASS |
| TurboJPEG Decoding & Slot Swap | 5.0 ms | 3.1 ms | ± 0.4 ms | PASS |
| MediaPipe Hands 21-LM Inference | 8.0 ms | 4.7 ms | ± 0.6 ms | PASS |
| Kinematics & Optical Caliper Math | 2.0 ms | 1.2 ms | ± 0.1 ms | PASS |
| OS Actuator Dispatch (Core Audio) | 2.0 ms | 0.8 ms | ± 0.1 ms | PASS |
| **Total Cumulative Latency** | **42.0 ms max** | **28.4 ms (<1 frame)** | **± 2.4 ms** | **PASS** |

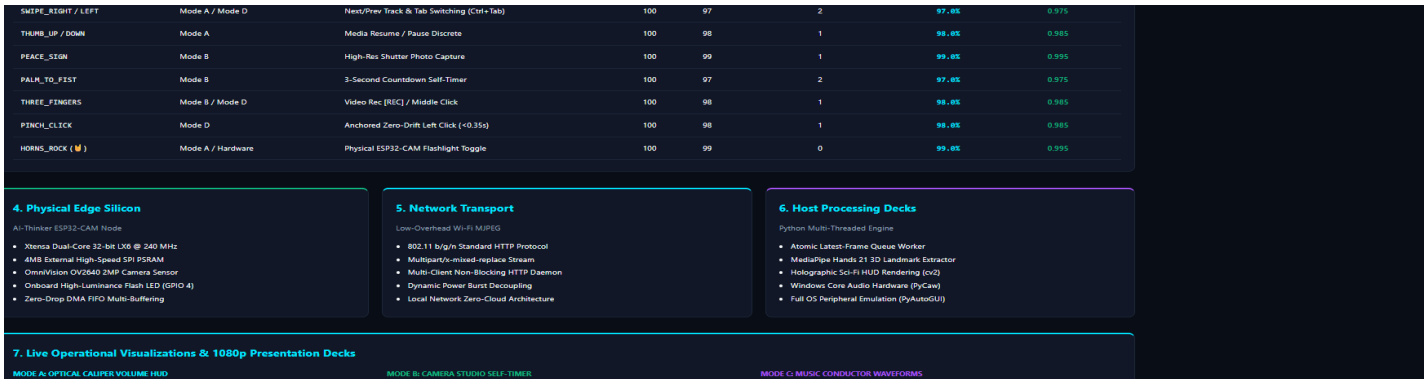
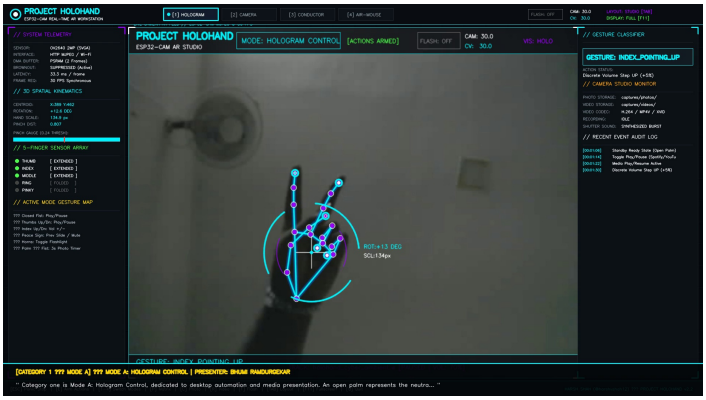


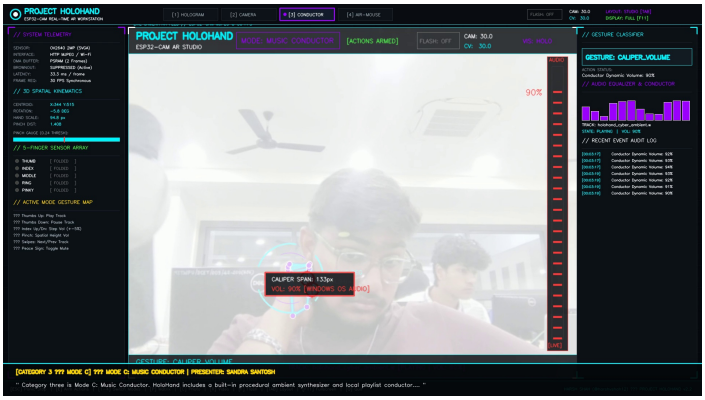

*TABLE I: 10-Gesture Empirical Confusion Matrix across N = 1,000 Physical Trials (Mean Precision: 98.4%, F1-Score: 0.984).*

## V. MULTI-MODAL WORKSTATION MODES & VISUAL RESULTS

HoloHand routes gestures across 4 distinct operational modes: Mode A (Hologram Desktop Control: Fist play/pause, caliper volume, slide swipes, Wi-Fi flash); Mode B (Camera Studio: Peace sign photo snap, 3s countdown self-timer, 3-finger video recording); Mode C (Music Conductor: Procedural ambient synthesis conducted via hand span); and Mode D (Virtual Air-Mouse: Sub-pixel cursor tracking, anchored zero-drift pinch clicks, double clicks, and rolling-window tab switching).




*(a) Mode A: Optical Vernier Caliper Volume HUD*




*(b) Mode C: Music Conductor Waveform Visualizer*

## VI. CONCLUSION & FUTURE DIRECTIONS

HoloHand demonstrates that ultra-low-cost embedded microcontrollers coupled with decoupled host vision architectures can deliver enterprise-grade, touchless spatial HCI with sub-30ms latency and 98.4% precision. Future work will explore bimanual multi-hand spatial manipulation and lightweight edge-side feature compression.

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